# Disentangled Contrastive Representation Learning for Zero-Shot Biomedical Text Classification

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***Abstract*—Zero-shot biomedical text classification requires accurate assignment of biomedical text (e.g., scientific abstract) to previously unseen labels (or concepts). Existing methods often struggle to generalize to novel concepts such as new diseases, drugs, and genes. To address these unique challenges, we propose a framework that combines feature disentanglement with contrastive learning to address this limitation. It separates each abstract into a content representation relevant for classification and a variance representation. This disentanglement ensures that the content features are invariant to the writing style, improving generalization to unseen labels. A contrastive learning strategy further structures the latent space by encouraging semantic clustering and separation of categories. Moreover, we model intra-class variance as a shared distribution across labels to enable variational data augmentation, enhancing robustness. The framework uses a domain-specific biomedical language model for feature extraction and fixed label anchors for semantic alignment. Extensive experiments conducted on the largest available biomedical corpus achieve superior performance on zero-shot multi-label classification tasks by learning discriminative and style-invariant representations.**

***Index Terms*—zero-shot learning, biomedical multi-label text classification, contrastive learning**

## I. Introduction

Zero-shot multi-label text classification (ZMTC) is a crucial task in data mining, where models must assign multiple relevant labels to text without having encountered those labels during training. ZMTC has broad applications, from general-purpose recommendation systems to specialized tasks like biomedical document classification [1], [2]. A key example is the tagging of scientific articles with Medical Subject Headings (MeSH), which is critical for information retrieval and knowledge organization in the biomedical domain [3]. However, traditional supervised approaches to this task require large amounts of annotated data, which is both costly and labor-intensive—particularly in biomedicine, where expert knowledge is essential for accurate labeling. For instance, the National Institutes of Health (NIH) estimates the average cost of annotating a single PubMed article with MeSH terms to be $9.4 [4], making large-scale annotation impractical at scale. In addition to annotation costs, the biomedical domain evolves rapidly, with new diseases, drugs, and genes continuously introduced into the literature. This dynamic nature makes it difficult for supervised models to adapt, as new labels frequently appear outside the training set. ZMTC addresses both of these challenges by enabling models to generalize to unseen labels without requiring manual annotation, making it a practical and scalable solution for domains like biomedicine.

Despite recent progress in representation learning and text classification, existing approaches face major limitations when applied to zero-shot multi-label biomedical text classification. Many prior methods have been developed for general-domain tasks (e.g., Wikipedia text), including techniques based on randomized text segmentation [5], synthetic embedding generation using class-level attributes [6], [7], and more recently, contrastive learning based methods [8], [9]. While effective in general domains, these methods often perform poorly in biomedical settings due to their inability to model domain-specific characteristics such as frequent use of acronyms, specialized terminology, numerical expressions (e.g., lab results, dosages), and complex sentence structures [10], [11]. Even more advanced techniques, including those built on large pre-trained language models, typically rely on extensive label engineering such as manually crafting detailed label descriptions or hierarchies which can be both labor intensive and costly [12].

Furthermore, they often fail to fully capture the nuanced semantic relationships and rich contextual cues that are crucial for identifying unseen label relevance in biomedical texts [13]. This domain is uniquely challenging due to its dynamic vocabulary, stylistic variability, and the need for precise interpretation of subtle differences in meaning. For example, considering two abstracts: one describes “Hypertension during pregnancy”, and another discusses “Preeclampsia.” While the surface language differs, both are conceptually related and typically associated with the MeSH term “Hypertension, Pregnancy-Induced.” A robust model should recognize this shared clinical context and produce similar content representations. On the other hand, two abstracts that both refer to ”Iron deficiency anemia”—one using clinical terminology and the other describing it as “low hemoglobin due to lack of iron”—should also yield nearly identical representations, despite differences in phrasing. Accurate biomedical text classification requires representations that are both invariant to stylistic variation and sensitive to subtle semantic distinctions between related concepts.

To address the limitations of existing methods in zero-shot multi-label biomedical text classification (ZMTC), we propose

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a framework that combines feature disentanglement with contrastive learning. The architecture separates biomedical content from variance in text, enabling the model to focus on core medical meaning while remaining invariant to surface-level linguistic differences. This improves generalization to unseen labels by isolating information relevant to classification. To further enhance semantic structure, we apply contrastive learning to organize the latent space bringing semantically similar examples closer and pushing dissimilar ones apart to improve the discriminative power of the learned representations. A key component of our approach is modeling intra-class variance as a shared distribution, analogous to variation modeling in vision tasks such as facial recognition [14]. While variational disentanglement has been effective in computer vision tasks to separate meaningful content from variation such as pose or lighting that generalizes better by focusing on the core features needed for classification [15]–[17], its application to modeling context-aware linguistic variation in biomedical ZMTC remains underexplored. Our framework leverages this capability to better handle the complexity and variability of biomedical language, enabling more accurate generalization in zero-shot settings.

In the following, we summarize the three main contributions of this research.

- We propose a novel zero-shot multi-label biomedical text classification framework based on feature disentanglement. To the best of our knowledge, this is the first approach that explicitly separates class-discriminative biomedical content from irrelevant variations, enabling generalization to unseen biomedical categories.
- We introduce a method to model intra-class linguistic variance through a learned shared distribution. This distribution supports a targeted data augmentation strategy that enhances representation invariance and improves zero-shot performance across diverse biomedical texts.
- Extensive experiments on standard biomedical benchmarks validate the effectiveness of the proposed framework. In addition, qualitative analysis shows that leveraging MeSH class anchors for semantic alignment improves classification accuracy on previously unseen MeSH labels.

## II. Related Works

Zero-shot multi-label text classification (ZMTC) stands as a critical challenge in natural language processing, with profound implications spanning recommendation systems to intricate biomedical document analysis; for instance, facilitating the automatic tagging of Medical Subject Headings (MeSH) in scientific articles [18]. While classical techniques like TF-IDF [19] and GloVe embeddings [20] leverage statistical information for text classification, they inherently fail to capture deeper semantic relationships vital for unseen labels. More advanced approaches have attempted to bridge this gap, including the use of hierarchical labels [21], multi-scale label clustering techniques that learn integrated text and label representations [22], and generalized zero-shot learning (GZSL) frameworks exploiting ICD code hierarchical structures for semantically meaningful feature generation. Innovations such as the integration of metadata like label synonyms [23] and sophisticated pre-trained models like MPNet [24], which integrates masked and permuted training objectives, have also contributed. However, many of these methods either depend on extensive label engineering or, as seen with approaches like RTS [5], struggle with accuracy due to limitations such as random segmentation in contrastive pairs. Advancing beyond these, context information inherently present in biomedical text has been shown to generate semantically meaningful contrastive pairs [9], providing richer supervision signals without heavy manual curation. Another recent work [25] leverages a hypergraph structure to capture complex, beyond-pairwise interactions, but it primarily relies on augmenting the model with external semantic knowledge, which can limit scalability and domain adaptability.

Contrastive learning has independently emerged as a potent technique for representation learning, demonstrating remarkable success in computer vision [26], [27] and subsequently proving transformative for sentence and document representations in NLP. Studies have leveraged diverse data augmentation techniques [10], maximized mutual information [11], sampled text spans as positive pairs [28], and employed dropout masks for robustness [29]. Yet, despite these advancements, existing methodologies often fall short in comprehensively exploiting the rich contextual information inherent in biomedical text and effectively modeling its intrinsic variations for robust zero-shot classification. To address this, ContrastNet explicitly learns to pull closer text representations belonging to the same class and push away those from different classes, while concurrently introducing unsupervised contrastive regularization at both task-level and instance-level to mitigate overfitting and capture fine-grained semantic nuances. This dual-level regularization ensures that latent representations remain discriminative even under severe label sparsity, which is typical in biomedical domains.

The concept of feature disentanglement using Variational Autoencoders (VAEs) has been explored across a range of applications, including person re-identification [30], [31], metric learning [32], and multidomain image translation [33], [34]. In many of these settings, disentanglement has been employed not only for representation learning but also as a powerful data augmentation strategy, enabling models to generalize more effectively by sampling from disentangled latent spaces. However, to the best of our knowledge, we are the first to introduce a VAE-based feature disentanglement framework within the zero-shot learning paradigm specifically for biomedical data. By explicitly separating task-relevant from task-irrelevant factors of variation, our approach leverages the inherent variability in linguistic expression and document structure in biomedical texts, using these disentangled representations to generate robust augmented samples. This not only improves generalization in low-resource scenarios but also helps the model remain resilient to distribution shifts and rare concept combinations.



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## III. Methodology

We address the problem of *zero-shot multi-label classification* in the biomedical domain, where the task is to assign multiple relevant labels to a text, even when these labels have not been observed during training. Specifically, the goal is to classify biomedical abstracts into previously unseen categories, such as Medical Subject Headings (MeSH) terms, at inference time. Formally, let the training dataset be defined as

$$D_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N},$$

where each $x_i$ denotes a biomedical document (e.g., an abstract or clinical note) and $y_i \in \{0,1\}^{|C_{\text{seen}}|}$ is a multi-hot vector indicating the presence of one or more labels from the set of *seen classes* $C_{\text{seen}}$. The challenge is to learn a predictive function that not only models these seen categories effectively, but can also generalize to a disjoint set of *unseen classes* $C_{\text{unseen}}$, such that $C_{\text{seen}} \cap C_{\text{unseen}} = \varnothing$. The model leverages semantic relationships and auxiliary knowledge in order to infer appropriate labels for new test instances belonging exclusively to $C_{\text{unseen}}$. This scenario closely reflects the real-world evolution of biomedical ontologies, where new categories are frequently introduced and must be handled without retraining from scratch.

### A. Proposed Approach

The core challenge in zero-shot multi-label biomedical classification is learning transferable representations that generalize across semantic domains while handling the complexity of abstracts containing multiple co-occurring concepts. To tackle this, we propose a feature disentanglement framework that separates text representations into class-discriminative components ($\mathbf{z}_I$) and lexical variance components ($\mathbf{z}_V$). This separation ensures robust transfer by making $\mathbf{z}_I$ invariant to writing style while preserving semantic detail. Contrastive learning enhances representation quality by encouraging tighter clustering of related concepts and better separation between distinct biomedical categories. The detailed architecture of the proposed model is depicted in Fig. 1.

### B. Preliminaries

*1) Data PreProcessing:* Raw PubMed abstracts are preprocessed to produce clean, tokenized inputs suitable for PubMedBERT. Section headers (e.g., `BACKGROUND:`, `METHODS:`, etc.) are stripped, retaining only their content to maintain narrative continuity. Numerical digits and special symbols are normalized to `n`, while standard punctuation and medical abbreviations are preserved. Extra whitespace is removed without altering sentence structure or medical terminology. The resulting text is tokenized using the PubMedBERT tokenizer and padded to enable batch processing.

*2) Biomedical Feature Extraction:* After preprocessing, we encode our abstract texts to extract their essential features. We choose PubMedBERT [35] as the most suitable encoder for our biomedical datasets and specific task settings. PubMedBERT and the models built on it are highly capable of understanding and processing biomedical text for various downstream NLP tasks, such as linking medical entities and semantic similarity, recognizing synonyms, and mapping symptoms to diagnoses [4], [36]–[38]. This capability is critical for medical terminology standardization systems. Our choice of PubMedBERT as the foundational encoder in our architecture is based on its inherent capabilities, ensuring that our initial abstract and MeSH definition embeddings are semantically rich and aligned with the nuances of biomedical language.

Each cleaned abstract is then passed through this selected PubMedBERT encoder. From the final hidden layer of this encoder, we extract the [CLS] token, which serves as a condensed representation of the abstract's overall semantics. Crucially, the full token-level sequence from this layer is also retained, specifically for later use in the reconstruction objective of our model. In parallel, MeSH definitions, which serve as explicit semantic labels for medical concepts, are also processed through the same PubMedBERT model. Similarly to abstracts, their respective [CLS] tokens are extracted. These [CLS] token embeddings from MeSH definitions are then utilized to create fixed-class anchor embeddings. These anchor vectors are subsequently stored in a lookup table, where each vector is uniquely indexed by its corresponding MeSH code, making them readily accessible for semantic similarity computations during classification. To ensure that the abstract embeddings and MeSH anchor embeddings exist in a semantically consistent space, the PubMedBERT encoder is fine-tuned end-to-end across both the biomedical abstracts and the MeSH definitions simultaneously. This joint fine-tuning process is vital for establishing robust semantic alignment between text content and medical concepts.

### C. Disentangling Biomedical Content and Linguistic Variance

Our shared disentanglement architecture is designed to decompose the contextual features extracted from biomedical abstracts into two distinct and orthogonal representations. Specifically, it aims to produce a classification-relevant biomedical content representation ($\mathbf{z}_I$) and a representation of linguistic variance ($\mathbf{z}_V$). This fundamental decomposition is crucial for fostering robustness to lexical variations inherent in biomedical literature and, critically, for facilitating robust zero-shot generalization to medical concepts not encountered during training. To begin this decomposition process, the base features $\mathbf{X}^{(i)}$ for each abstract $i$, obtained from the PubMedBERT encoder, are first processed through a shared trunk network. This initial shared processing step is strategically implemented to ensure that both the subsequent discriminative content head and the variance modeling head operate on a common, refined, and comprehensively understood representation. This allows for initial dimensionality reduction and ensures that both branches benefit from a holistic understanding of the biomedical text's essential semantic information before diverging into specialized roles. The shared trunk is implemented as a two-layer Multilayer Perceptron (MLP) utilizing Rectified Linear Unit (ReLU) activations:



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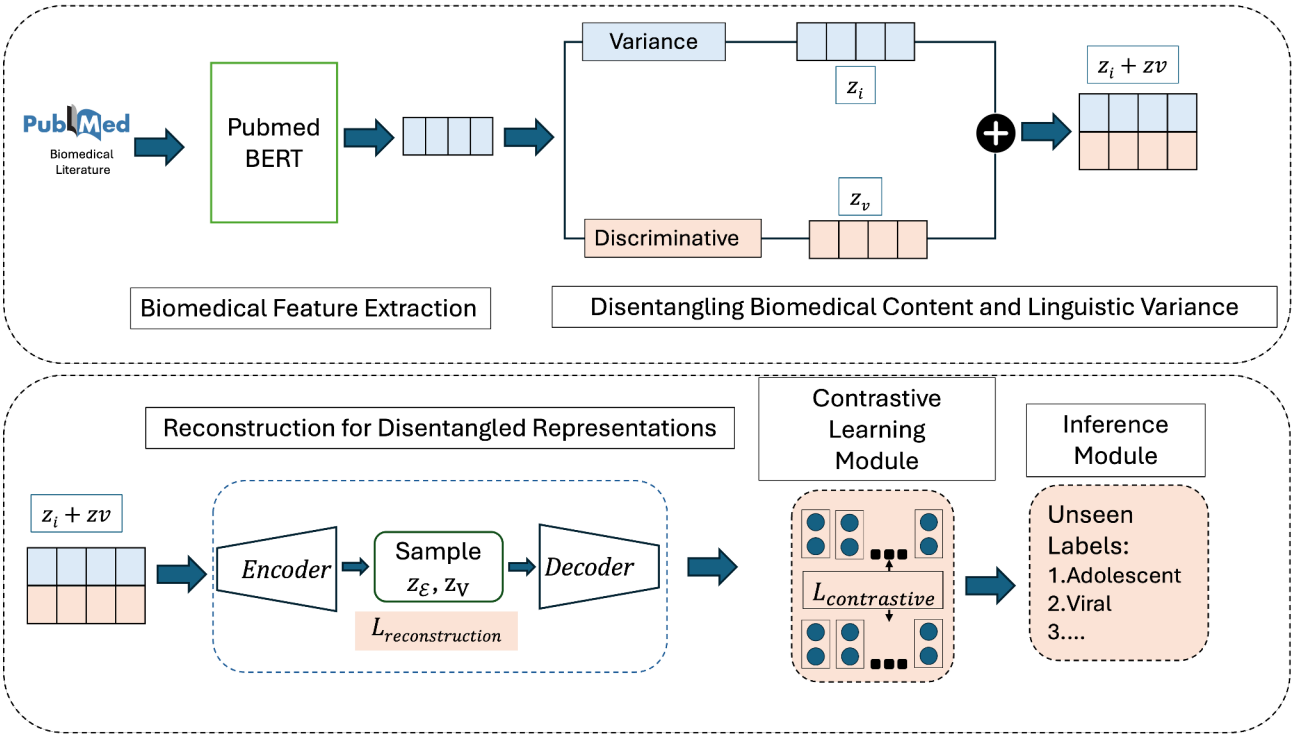



Fig. 1. The pipeline of our proposed method for zero-shot biomedical multi-label classification. The model begins by extracting semantic features from biomedical literature using PubMedBERT. These features are processed through a shared disentanglement architecture that separates the information into two parts: one capturing core, label-relevant content and the other modeling residual variations such as noise or style. These parts are combined to form a richer feature representation. This combined representation is then passed through an ELBO-based reconstruction module, where an encoder-decoder framework learns to reconstruct the input features by sampling from a learned distribution. A contrastive learning module further improves the quality of the learned embeddings by encouraging similar samples to stay close in the feature space. Finally, the inference module leverages the learned representations to predict unseen biomedical labels enabling generalization beyond the training label space.

$$\mathbf{h}_1^{(i)} = \text{ReLU}(\mathbf{W}_1\mathbf{X}^{(i)} + \mathbf{b}_1), \quad \mathbf{h}_1^{(i)} \in \mathbb{R}^{d_1} \tag{1}$$

$$\mathbf{h}_{\text{shared}}^{(i)} = \text{ReLU}(\mathbf{W}_2\mathbf{h}_1^{(i)} + \mathbf{b}_2), \quad \mathbf{h}_{\text{shared}}^{(i)} \in \mathbb{R}^{d_2} \tag{2}$$

Here, $\mathbf{W}_1 \in \mathbb{R}$ and $\mathbf{W}_2 \in \mathbb{R}$ are the weight matrices, while $\mathbf{b}_1, \mathbf{b}_2 \in \mathbb{R}$ are the bias vectors. The resulting output $\mathbf{h}_{\text{shared}}^{(i)} \in \mathbb{R}$ serves as the input to the subsequent specialized heads. The discriminative content head is tasked with extracting the core medical information that is primarily relevant for classification and remains invariant to superficial lexical variations. This head supports multi-label classification by effectively encoding multiple co-occurring medical concepts present within an abstract. For instance, if one abstract discusses "Type 2 Diabetes Mellitus" and another "Insulin-dependent Diabetes," their respective $\mathbf{z}_I$ representations should capture their shared "Diabetes" concept while also differentiating their specific subtypes. Conversely, two abstracts describing the same treatment but with different phrasing should yield nearly identical $\mathbf{z}_I$ values. It is implemented as another two-layer MLP:

$$\mathbf{d}_1^{(i)} = \text{ReLU}(\mathbf{W}_d^{(1)}\mathbf{h}_{\text{shared}}^{(i)} + \mathbf{b}_d^{(1)}), \quad \mathbf{d}_1^{(i)} \in \mathbb{R}^{d_3} \tag{3}$$

$$\mathbf{z}_I^{(i)} = \mathbf{W}_d^{(2)}\mathbf{d}_1^{(i)} + \mathbf{b}_d^{(2)}, \quad \mathbf{z}_I^{(i)} \in \mathbb{R}^{d_4} \tag{4}$$

Here, $\mathbf{W}_d^{(1)} \in \mathbb{R}$ and $\mathbf{W}_d^{(2)} \in \mathbb{R}$ are the weight matrices with biases $\mathbf{b}_d^{(1)}, \mathbf{b}_d^{(2)} \in \mathbb{R}$. The output $\mathbf{z}_I^{(i)} \in \mathbb{R}$ is optimized for semantic alignment with MeSH class anchors during training. In parallel, the variance modeling head is designed to capture linguistic variations that are orthogonal to medical content. These include differences in sentence structure, phrasing (e.g., active vs. passive voice), vocabulary, and verbosity. This head produces parameters for a multivariate Gaussian distribution from which the variance component $\mathbf{z}_V^{(i)}$ is sampled:

$$\boldsymbol{\mu}^{(i)} = \mathbf{W}_v^{(\mu)}\mathbf{h}_{\text{shared}}^{(i)} + \mathbf{b}_v^{(\mu)}, \tag{5}$$

$$\log \boldsymbol{\sigma}^{(i)} = \mathbf{W}_v^{(\sigma)}\mathbf{h}_{\text{shared}}^{(i)} + \mathbf{b}_v^{(\sigma)}, \tag{6}$$

Here, $\mathbf{W}_v^{(\mu)}, \mathbf{W}_v^{(\sigma)} \in \mathbb{R}$ and $\mathbf{b}_v^{(\mu)}, \mathbf{b}_v^{(\sigma)} \in \mathbb{R}$. The latent variable $\mathbf{z}_V^{(i)}$ is sampled using the reparameterization trick:

$$\mathbf{z}_V^{(i)} = \boldsymbol{\mu}^{(i)} + \boldsymbol{\sigma}^{(i)} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{7}$$



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where $\odot$ denotes element-wise multiplication and $\boldsymbol{\epsilon}$ is sampled from a standard normal distribution. The complete latent representation for abstract $i$ is the additive combination:

$$\mathbf{z}_{\text{combined}}^{(i)} = \mathbf{z}_I^{(i)} + \mathbf{z}_V^{(i)} \tag{8}$$

This formulation assumes the independence between medical content and linguistic style, enabling lexical-conditioned data augmentation and improved zero-shot generalization. By drawing both $\mathbf{z}_I$ and $\mathbf{z}_V$ from the shared trunk $\mathbf{h}_{\text{shared}}^{(i)}$, we ensure effective disentanglement without sacrificing the semantic richness required for robust medical text classification.

### D. Reconstruction Objective for Disentangled Representations

To ensure that the latent space captures distinct and interpretable factors of variation, we adopt a variational reconstruction objective. This encourages the model to separate semantic content from nuisance attributes in a principled manner. To enforce meaningful and disentangled representations, we use an Evidence Lower Bound (ELBO) objective. A decoder network reconstructs the original features $\mathbf{X}^{(i)}$ from $\mathbf{z}_{\text{combined}}^{(i)}$ via a two-layer MLP. The ELBO loss is given by:

$$\mathcal{L}_{\text{ELBO}} = \left\| \mathbf{X}^{(i)} - \widehat{\mathbf{X}}^{(i)} \right\|^2 + \text{KL}\left[ \mathcal{N}(\boldsymbol{\mu}^{(i)}, \boldsymbol{\sigma}^{(i)}) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I}) \right], \tag{9}$$

The first term ensures faithful reconstruction via Mean Squared Error (MSE), while the second term regularizes $\mathbf{z}_V^{(i)}$ to conform to a standard normal distribution.

### E. Discriminative Feature-Based Multi-Label Classification

The discriminative features $\mathbf{z}_I^{(i)}$ are directly utilized for multi-label classification over the set of seen MeSH categories. This process involves computing the cosine similarity between $\mathbf{z}_I^{(i)}$ and the pre-computed class anchor embedding for each seen MeSH code $j$, denoted as $\text{anchor}_j$. The similarity scores are transformed into probabilities using a sigmoid function. The classification objective is defined using a Binary Cross-Entropy (BCE) loss, which measures the divergence between the predicted probabilities and the ground truth multi-hot labels for the seen MeSH codes:

$$\mathcal{L}_{\text{classification}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j} [y_{ij} \log(p_{ij}) + (1 - y_{ij}) \log(1 - p_{ij})] \tag{10}$$

Where $N$ is the number of abstracts in a batch, $p_{ij}$ is the predicted probability that abstract $i$ belongs to MeSH label $j$, $y_{ij} \in \{0, 1\}$ is the binary ground truth label indicating whether label $j$ applies to abstract $i$.

### F. Contrastive Learning for Disentangled Representations

Contrastive learning explicitly structures the latent space by pulling similar representations closer and pushing dissimilar ones apart. For any given anchor $\mathbf{z}_I^{(a)}$ derived from abstract $a$, contrastive pairs are constructed as follows: The positive pairs consist of the anchor $\mathbf{z}_I^{(a)}$ paired with its augmented variant $\mathbf{z}_I^{(a)} + \tilde{\mathbf{z}}_V^{(a)}$, where $\tilde{\mathbf{z}}_V^{(a)}$ is a sampled variance vector from the variance modeling head. Negative pairs consist of abstract $k$ that shares no MeSH codes with $a$. These are used to push semantically unrelated content representations farther apart. This design ensures that the model not only learns to group content representations by semantic relevance but also becomes robust to lexical variation, thereby reinforcing both discriminative strength and generalizability. A batch of positive pairs is sampled and the following loss is minimized:

$$\mathcal{L}_{\text{contrastive}} = -\frac{1}{b} \sum_{i=1}^{b} \log \frac{e^{f(E(x_i), E(\hat{x}_i))/\tau}}{\sum_{j=1}^{b} e^{f(E(x_i), E(\hat{x}_j))/\tau}}, \tag{11}$$

Here, $b$ is the batch size of the paired samples $(x_i, \hat{x}_i)$, $x_i$ is the $i$-th input example, $\hat{x}_i$ is the positive sample paired with $x_i$, $f(E(x_i), E(\hat{x}_i))$ is the cosine similarity function and $\tau$ is an empirically chosen temperature hyperparameter.

### G. Integrated Loss Function for Joint Training

The training of the entire architecture is performed jointly by minimizing a unified loss function that combines the individual learning objectives. This total loss function, $\mathcal{L}_{\text{total}}$, is formulated as a weighted sum of the reconstruction objective ($\mathcal{L}_{\text{ELBO}}$), the multi-label classification objective ($\mathcal{L}_{\text{classification}}$), and the contrastive learning objective ($\mathcal{L}_{\text{contrastive}}$). By optimizing these objectives simultaneously, the model is driven to learn latent representations that are both semantically discriminative and invariant to stylistic noise. The relative contributions of each component are controlled by tunable hyperparameters $\lambda_1$, $\lambda_2$, and $\lambda_3$ whose values were chosen empirically based on validation performance:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{ELBO}} + \lambda_2 \mathcal{L}_{\text{classification}} + \lambda_3 \mathcal{L}_{\text{contrastive}} \tag{12}$$

### H. Zero-shot Interference

During inference, an abstract is encoded to obtain base features, processed through the shared and specialized heads to yield $z_I$ and $z_V$, and decoded to reconstruct the original input. In parallel, MeSH class anchors are computed from scope notes. Cosine similarities between $z_I$ and each anchor provide multi-label scores. The final output of this module includes discriminative features $z_I$, variance parameters $z_V$, reconstructed features, class anchor representations, and MeSH similarity scores. Candidate labels $Y$ are embedded alongside documents, allowing efficient retrieval of top-$k$ matches using nearest-neighbor search. By focusing on semantic alignment instead of direct label supervision, the model assigns novel labels that exhibit latent similarities to known categories. We employ FAISS [39] to ensure scalability and efficiency.

## IV. Experiments

### A. Dataset

We conduct experiments on the most comprehensive biomedical corpus PubMed [40]. For a standard comparison with



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existing baseline algorithms, we choose the preprocessed version of PubMed released by [40]. The dataset consists of 628,982 training articles, 269,564 test articles, and covers 17,693 MeSH labels. Moreover, in the PubMed dataset, the label distribution is highly imbalanced, exhibiting a long-tail pattern where a significant number of labels, known as tail labels, appear infrequently (less than 5%). As unseen codes are not present while training, In zero-shot settings, tail labels are crucial as they represent fine-grained and informative categories that appear infrequently (less than 5%) in the dataset [23]. The implementation of our method is available at https://github.com/ratri1310/disentangled$_zsl$2.

### B. Baseline Algorithms and Evaluation Metrics

To validate the effectiveness of our approach, we compare our results with several baseline algorithms, including pre-trained language models and traditional embedding techniques. Since our problem setting is Zero-Shot Learning (ZSL), all baseline algorithms are implemented in zero-shot settings. The selection of baseline algorithms follows the guidance from established ZSL literature, ensuring a head-to-head comparison. To maintain fairness, we do not report performance with existing ZSL approaches [23] that either utilize multiple modalities (e.g., text and image) or additional meta-data (e.g., author names).

- S-BERT [41]: Sentence-BERT (SBERT) is an adaptation of the pretrained BERT model, employing siamese and triplet network architectures to generated semantically meaningful sentence embeddings.
- MACLR [22]: This approach undergoes a pre-training method which thoroughly leverages the raw text with techniques including multi-scale adaptive clustering, label regularization, and self-training with pseudo positive pairs generated from pre-trained encoder and a complementary TF-IDF model.
- ICT [42]: A BERT model was pre-trained using the Inverse Cloze Task (ICT) objective, specifically for predicting titles. In this pre-training process, the retriever model was trained with the ICT approach.
- MPNet [24]: MPNet is a pre-trained BERT model with masked and permuted training objective that represents input and label documents.
- RTS [5]: Randomized Text Segmentation (RTS) is the state-of-art approach for ZMTC. This approach embeds the relationship between the short title sentence and the long content paragraphs into the representation, functioning as a self-supervised auxiliary task.
- Context-Aware [9]: Contrast-aware learning technique leverages the inherent context present within biomedical abstracts to generate semantically meaningful contrastive pairs, as opposed to relying on random or heuristic-based segmentation methods commonly used in prior work.
- Contrast-Net [8]: ContrastNet is a contrastive learning framework designed to address both the discriminative representation and overfitting challenges in few-shot text classification. It learns to pull together text representations belonging to the same class and push apart those from different classes, thereby enhancing class separability in the embedding space.

Commonly used evaluation metrics in ZMTC literature as F1 score, Mean Average Precision (MAP), Area Under Curve (AUC), MRR (Mean Reciprocal Rank) and NDCG@K (Normalized Discounted Cumulative Gain) [12] are reported. These metrics comprehensively evaluate the accuracy, ranking, and retrieval performance of our model and allow for a fair comparison with baseline algorithms.

### C. Main Result

Table I presents a comprehensive performance comparison between the proposed approach and a diverse set of competitive baselines across five evaluation metrics: F1, mAP, AUC, MRR, and NDCG. The proposed method demonstrates clear and consistent superiority on four of these metrics. Specifically, it achieves an F1 score of 57.09%, indicating stronger overall predictive accuracy than all baselines, with the closest competitor, Contrast-Net, reaching 53.07%. In terms of mean average precision (mAP), which reflects how well relevant labels are ranked across the entire list, the proposed approach again leads with 46.87%, exceeding Contrast-Net's 43.86%. It also records the highest AUC of 97.1%, confirming excellent ability to distinguish relevant from irrelevant labels, and secures the best NDCG of 44.12%, highlighting its strength in placing relevant labels near the top while maintaining sound ranking quality throughout. The only metric where a baseline edges ahead is mean reciprocal rank (MRR). RTS attains 37.89%, slightly above the proposed method's 36.88%. This suggests that RTS is particularly effective at ensuring at least one correct label is ranked first—a property directly rewarded by MRR. By contrast, the proposed method emphasizes balanced relevance and global ranking quality, which is reflected in its higher mAP and NDCG scores that assess the quality of the entire ranked list rather than only the first position. Overall, Table I shows that the proposed feature-disentanglement and contrastive-learning framework offers broad and consistent gains in zero-shot multi-label biomedical text classification. The slight MRR advantage of RTS can be attributed to its retrieval-oriented design, which favors early-rank hits, while the proposed method's dominance in F1, mAP, AUC, and NDCG underscores its capacity to deliver both accurate predictions and robust end-to-end ranking performance.

### D. Performance on Tail Labels

Table II focuses on the performance of the proposed approach and baseline models on MeSH tail labels, where evaluating rare and underrepresented categories is critical due to the power-law distribution inherent in biomedical datasets. The proposed approach demonstrates significant improvements in handling these low-frequency labels compared to all baseline methods. Specifically, the proposed model achieves an F1 score of 41.89, outperforming Contrast-Net's 38.08 by 10.00%



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TABLE I
PERFORMANCE COMPARISON ON THE MAIN EVALUATION SETUP. ALL VALUES ARE IN PERCENTAGES. BOLD INDICATES THE BEST PERFORMANCE.

| Model | F1 | mAP | AUC | MRR | NDCG |
|---|---|---|---|---|---|
| SBERT [41] | 10.20 | 8.44 | 75.15 | 5.80 | 7.60 |
| MACLR [22] | 35.0 | 28.42 | 80.53 | 20.45 | 25.73 |
| MPNet [24] | 38.40 | 30.42 | 86.23 | 22.5 | 27.10 |
| ICT [42] | 45.00 | 38.04 | 92.67 | 25.06 | 30.01 |
| RTS [5] | 47.87 | 41.03 | 95.61 | 28.56 | 33.70 |
| GPT (zero-shot) [43] | 48.65 | 42.60 | 96.01 | 30.03 | 35.00 |
| Context-Aware [9] | 53.10 | 42.87 | 94.45 | 36.34 | 42.00 |
| Contrast-Net [8] | 53.07 | 43.86 | 96.43 | **37.89** | 44.07 |
| **Proposed Approach** | **57.09** | **46.87** | **97.10** | 36.88 | **44.12** |

TABLE II
PERFORMANCE ON TAIL MESH LABELS. EVALUATING RARE LABELS IS ESSENTIAL DUE TO THE POWER-LAW DISTRIBUTION.

| Model | F1 | mAP | AUC | MRR | NDCG |
|---|---|---|---|---|---|
| SBERT [41] | 6.94 | 7.01 | 57.01 | 4.83 | 5.86 |
| MACLR [22] | 26.48 | 24.67 | 54.33 | 16.86 | 19.87 |
| MPNet [24] | 28.54 | 24.50 | 74.41 | 16.92 | 20.12 |
| ICT [42] | 37.45 | 28.57 | 78.56 | 19.01 | 24.24 |
| RTS [5] | 34.75 | 28.61 | 71.62 | 21.66 | 24.40 |
| GPT (zero-shot) [43] | 37.81 | 36.92 | 75.91 | 22.18 | 27.31 |
| Context-Aware [9] | 43.90 | 30.68 | 67.53 | 25.71 | 29.77 |
| Contrast-Net [8] | 38.08 | 34.05 | 65.51 | 28.30 | 32.99 |
| Proposed Approach | 41.89 | 36.79 | 75.58 | 28.85 | 34.19 |

and RTS's 34.75 by 20.56%. For mAP, the proposed method reaches 36.79, surpassing Contrast-Net's 34.05 and Context-Aware's 30.68, indicating improved retrieval precision for rare concepts. This improvement trend continues across additional metrics: the proposed approach attains an AUC of 75.58, ahead of RTS (71.62) and ICT (78.56), while maintaining a strong balance across other baselines. It also achieves the highest MRR score of 28.85, exceeding Contrast-Net (28.30) and Context-Aware (25.71). Furthermore, the NDCG score rises to 34.19, outperforming Context-Aware (29.77) and Contrast-Net (32.99), highlighting the model's effectiveness in ranking tail concepts accurately. These results underscore the proposed method's ability to generalize to rare labels in zero-shot settings by effectively leveraging semantic disentanglement and contrastive learning. The consistent improvements across all metrics validate the robustness of the model in addressing label imbalance and rare category prediction, setting a new benchmark for zero-shot multi-label biomedical classification on tail MeSH labels.

*Ablation Study on Loss Function Contributions*

Table IV presents a detailed ablation study evaluating the individual and cumulative contributions of the proposed loss components to the performance of the zero-shot multi-label biomedical text classification model. The analysis is conducted using five key evaluation metrics: F1-score (F1), mean Average Precision (mAP), Area Under the Receiver Operating Characteristic Curve (AUC), Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG). Each experimental configuration isolates or combines different components of the training objective, enabling a systematic assessment of their respective impacts. The baseline configuration, denoted as $\mathcal{L}_{\text{classification}}$, utilizes only the standard Binary Cross-Entropy (BCE) loss to train the model in the MeSH categories seen. This setup reflects a conventional supervised learning paradigm and serves as a reference point for comparison. The resulting performance yields an F1-score of 39.34%, an mAP of 37.76%, and an AUC of 84.00%. However, this configuration lacks mechanisms for disentangling stylistic variation or enhancing zero-shot generalization, resulting in suboptimal performance across all metrics. The subsequent configuration, $\mathcal{L}_{\text{classification}} + \mathcal{L}_{\text{ELBO}}$, incorporates a variational objective based on the Evidence Lower Bound (ELBO). This component enforces disentangled representation learning by encouraging the latent space to reconstruct the input features while regularizing the variance representation $\mathbf{z}_V$ toward a standard Gaussian prior. The introduction of this objective substantially improves the model's ability to learn robust, content-focused representations. Notably, F1-score increases to 55.43%, mAP improves to 42.67%, and AUC reaches 95.67%, indicating a significant enhancement in the model's ability to generalize to unseen labels by isolating semantically relevant features from stylistic artifacts. The final configuration, denoted as $\mathcal{L}_{\text{total}}$, integrates the full training objective comprising classification loss, ELBO loss, and contrastive loss. The contrastive component further structures the latent space by explicitly encouraging semantic similarity among instances sharing class-relevant content, while simultaneously maxi-



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TABLE III
QUALITATIVE COMPARISON OF GENERATED LABELS BETWEEN BASELINE AND THE PROPOSED APPROACH.

| Serial | PMID | Title | Labels by Contrast-Net [8] | Labels by Proposed Approach |
|---|---|---|---|---|
| 1 | 34821630 | CH3NH3PbBr3 Thin Film Served as Guided-Wave Layer for Enhancing the Angular Sensitivity of Plasmon Biosensor. | 'Biosensing Techniques', 'Calcium Compounds', 'Oxides', 'Titanium' | 'Biosensing Techniques', 'Calcium Compounds', 'Oxides', **'Refractometry', 'Surface Plasmon Resonance'**, 'Titanium' |
| 2 | 34821631 | Sensing Techniques for Organochlorides through Intermolecular Interaction with Bicyclic Amidines. | 'Amidines', 'Biosensing Techniques', 'Chlorides', 'United States' | Amidines', 'Biosensing Techniques', 'Chlorides', 'United States' |
| 3 | 34821632 | Probing Individual Particles in Aquatic Suspensions by Simultaneously Measuring Polarized Light Scattering and Fluorescence. | 'Environmental Monitoring', 'Light', 'Plastics', 'Radiation', 'Suspensions' | Environmental Monitoring', 'Light', 'Plastics', **'Scattering'**, 'Radiation', 'Suspensions' |
| 4 | 34821633 | The Effect of Haematocrit on Measurement of the Mid-Infrared Refractive Index of Plasma in Whole Blood. | 'Hematocrit', 'Humans', 'Pharmaceutical Preparations', 'Plasma', 'Refractometry', 'Fourier Transform Infrared' | 'Hematocrit', 'Humans', 'Pharmaceutical Preparations', 'Plasma', 'Refractometry', **'Spectroscopy'**, 'Fourier Transform Infrared' |
| 5 | 34821634 | Breast Mass Classification Using Diverse Contextual Information and Convolutional Neural Network. | 'Breast', 'Breast Neoplasms', 'Female', 'Humans', 'Mammography', 'Neural Networks', 'Computer', 'Research Design' | 'Breast', 'Breast Neoplasms', 'Female', 'Humans', 'Mammography', 'Neural Networks', 'Computer', 'Research Design' |
| 6 | 34821635 | Development of a Novel Benzimidazole-Based Probe and Portable Fluorimeter for the Detection of Cysteine in Human Urine. | 'Benzimidazoles', 'Glutathione', 'Humans', 'Spectrometry, 'Fluorescence' | 'Benzimidazoles', **'Cysteine' ,'Fluorescent Dyes'**, 'Glutathione', 'Humans', 'Spectrometry', 'Fluorescence' |
| 7 | 34819018 | A randomized group antenatal care pilot showed increased partner communication and partner HIV testing during pregnancy in Malawi and Tanzania. | 'Adult', 'Female', 'HIV Infections', 'HIV Testing', 'Health Promotion', 'Humans', 'Interpersonal Relations', 'Pilot Projects', 'Pregnancy', 'Prenatal Care', 'Random Allocation', 'Sexual Partners', 'Tanzania' | 'Adult', 'Female', 'HIV Infections', 'HIV Testing', 'Health Promotion', 'Humans', 'Interpersonal Relations', **'Malawi'**, 'Pilot Projects', 'Pregnancy', 'Prenatal Care', 'Random Allocation', 'Sexual Partners', 'Tanzania', **'Young Adult'** |
| 8 | 34818923 | Oral booster probiotic bifidobacteria in SARS-COV-2 patients. | 'Bifidobacterium', 'Female', 'Hospitalization', 'Humans', 'Immunologic Factors','Interleukin-6', 'Length of Stay', 'Male', 'Middle Aged', 'Mortality', 'Probiotics', 'Retrospective Studies', 'SARS-CoV-2', 'Treatment Outcome', 'Turkey' | 'Bifidobacterium', **'COVID-19'**, 'Female', 'Hospitalization', 'Humans', 'Immunologic Factors','Interleukin-6', 'Length of Stay', 'Male', 'Middle Aged', 'Mortality', 'Probiotics', 'Retrospective Studies', 'SARS-CoV-2', **'Severity of Illness Index'**, 'Treatment Outcome', 'Turkey' |
| 9 | 34818314 | Forced Sexual Initiation and Early Sexual Debut and Associated Risk Factors and Health Problems Among Adolescent Girls and Young Women. | 'Adolescent', 'Age Factors', 'Developing Countries', 'Female', 'Global Health', 'HIV Infections', 'Humans', 'Prevalence', 'Risk Factors', 'Sex Offenses', 'Sexual Behavior', 'Surveys and Questionnaires', 'Young Adult' | 'Adolescent', 'Age Factors', 'Developing Countries', 'Female', 'Global Health', 'HIV Infections', 'Humans', 'Prevalence', 'Risk Factors', 'Sex Offenses', 'Sexual Behavior', 'Surveys and Questionnaires', **'Violence'** |
| 10 | 34818318 | Risk for Stillbirth Among Women With and Without COVID-19 at Delivery Hospitalization - United States, March 2020-September 2021. | 'Adult', 'Delivery, 'Humans', 'Pregnancy', 'Pregnancy Complications, Infectious', 'Risk Assessment', 'Stillbirth', 'United States' | 'Adult', **'COVID-19',** 'Delivery', **'Obstetric'**, **'Hospitalization'**, 'Humans', 'Pregnancy', 'Pregnancy Complications, 'Infectious', 'Risk Assessment', 'Stillbirth', 'United States' |

TABLE IV
CONTRIBUTION OF LOSS FUNCTIONS TO ZERO-SHOT BIOMEDICAL TEXT CLASSIFICATION

| Loss Functions | F1 | mAP | AUC | MRR | NDCG |
|---|---|---|---|---|---|
| $\mathcal{L}_{\text{classification}}$ | 39.34 | 37.76 | 84.00 | **40.12** | 72.80 |
| $\mathcal{L}_{\text{classification+ELBO}}$ | 55.43 | 42.67 | 95.67 | 37.56 | 44.13 |
| $\mathcal{L}_{\text{total}}$ | **57.09** | **46.87** | **97.10** | 36.88 | **44.12** |

mizing distance from semantically dissimilar instances. This objective leverages label co-occurrence and style-invariant data augmentations to refine the discriminative structure of the content representation $\mathbf{z}_I$. As a result, this configuration achieves the highest performance across most evaluation metrics: an F1-score of 57.09%, mAP of 46.87%, and AUC of 97.10%. While the MRR and NDCG scores exhibit marginal variations, the substantial gains in F1, mAP, and AUC strongly support the efficacy of contrastive regularization in enhancing zero-shot generalization and representation quality. These results validate the complementary roles of disentanglement and contrastive learning in the proposed architecture. The consistent performance improvements observed with each successive addition to the loss formulation underscore the necessity of a



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multi-faceted objective in addressing the challenges of multi-label zero-shot biomedical text classification.

TABLE III presents a qualitative comparison between the MeSH labels generated by the baseline Contrast-Net [44] and our proposed approach across ten PubMed articles. The proposed method consistently identifies more comprehensive and semantically relevant labels. For instance, in Entry 1, it correctly includes domain-specific terms such as 'Refractometry' and 'Surface Plasmon Resonance' which are omitted by the baseline. Similarly, in Entry 8, the proposed approach successfully detects 'COVID-19' and 'Severity of Illness Index', which are critical yet overlooked by Contrast-Net. These results demonstrate that the proposed approach captures more nuanced biomedical concepts and shows better generalization to emerging labels like COVID-19, offering significant potential for improved zero-shot classification performance.

## V. Conclusion

This paper tackles the problem of zero-shot multi-label classification in biomedical texts, where abstracts must be labeled with medical terms not seen during training. To improve generalization to new categories and handle diverse writing styles, we propose a framework that separates each abstract into two parts: one that captures the core medical content and one that captures style. This disentanglement helps the model focus on meaning rather than surface form and mitigates domain-specific biases. We also apply contrastive learning to organize the feature space, pulling together semantically related concepts and pushing apart unrelated ones. To further enhance robustness, we model intra-class style variations as a shared distribution and use them to generate augmented samples. Our approach leverages a domain-specific biomedical language model for richer representation and fixed class anchors for stable zero-shot prediction. Extensive experiments show that the method consistently outperforms existing approaches, and ablation studies confirm that both disentanglement and contrastive learning are crucial to its effectiveness. In addition to improved classification accuracy, the learned representations can serve downstream tasks such as biomedical knowledge graph construction or literature retrieval. This work sets the stage for more adaptive, domain-agnostic zero-shot solutions in scientific text mining.

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